SENTEC | THE ENGINEER'S CODE

THE WEB FORCES

LogiPulse:
Courier & Last-Mile Delivery Platform

Danish | Module Head | 3rd Year Software Engineering

Before We Begin

A quick show of hands

1. Who here has participated in a hackathon before?
2. Who has built a full-stack web application before?
3. Who thinks 2 hours is enough to build a courier platform?

Your Mission

You are not just participants. You are the founding engineers of LogiPulse.

1

Request

A customer requests a delivery.

2

Deliver

A nearby rider accepts and delivers it.

3

Connect

Customer, rider and operations team stay connected through one shared system.

The Problem Statement

Build one connected courier workflow across three role-based dashboards.

USER

- Request courier
- Track own orders
- Assigned rider + chat

RIDER

- See available orders
- Find nearby pickups
- Accept + update status

ADMIN

- Manage platform
- Monitor orders / users / riders
- View statistics

One Order. One Connected Flow.

The dashboards must share data. Disconnected screens are not the product.

1

User creates request

Status: Pending

2

Request appears

Admin + Riders

3

Rider accepts

Finds nearby request

4

User sees rider

Assigned rider shown

5

User and Rider chat

Stay in touch

6

Rider updates status

Picked Up > In Transit > Delivered

7

Everyone sees updates

User + Admin

Shared data, one live system.

The Feature That Makes It Interesting

SMART NEARBY PICKUP

Order #21 1.4 km



Order #08 3.2 km



Order #15 5.8 km



Sorted nearest first

Real logic

Distance must be calculated, not typed in.

Valid approaches

Coordinates + distance formula, or predefined areas.

Not accepted

Hard-coded "1.2 km away" labels.

Build Smart, Not Overcomplicated

Focus on a working product, not unnecessary infrastructure.

ENCOURAGED

Use a relational database

Store users, riders and orders in related tables. It keeps your data consistent across all three dashboards.

PostgreSQL | MySQL | SQLite | Supabase

KEEP IT SIMPLE

- A complex backend is NOT mandatory
- Node/Express, Firebase or LocalStorage are acceptable
- Any modern frontend stack is allowed

Day 1: 120 Minutes

Core Product Engineering

00-15

minutes

Briefing + architecture

15-60

minutes

Three dashboards

60-100

minutes

Complete courier flow

100-120

minutes

Nearby pickup + chat

What You Must Submit

Build it, deploy it, explain it.

1	**Live web application**	Fully functional and deployed
2	**Public GitHub repository**	With clean commits
3	**README**	Overview, stack, 3-dashboard architecture, nearby-pickup logic
4	**Submission form**	Team + GitHub + live URL + members
5	**3-minute live demo**	Show the product working

How You Win

100 points. Functionality matters.



And One More Thing

Your system should be easy to MODIFY.

Day 2 may test your architecture, not just your coding speed.

BUILD. CONNECT. DELIVER.

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Questions before the clock starts?